

DEVANSH SHARMA

Applied AI Systems Architect | Autonomous Agents & OS Runtimes

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EXECUTIVE SUMMARY

Systems-focused AI Software Engineer specializing in autonomous multi-agent orchestration, cognitive persistent memory, and low-latency desktop/edge execution. Proven record architecting multi-tiered agent swarms, tri-hybrid retrieval engines (Reciprocal Rank Fusion), and multiprocessing pipelines eliminating Python GIL bottlenecks. Grand Champion at the NeuroLogic '26 Global NLP Datathon; dedicated to zero-idle telemetry, hardware-aware execution, and production-grade software engineering.

TECHNICAL SKILLS MATRIX

Languages & Core: Python, C++, TypeScript, SQL, Bash, HTML5 Canvas API

Applied AI & Agents: Multi-Agent Swarms (Supervisor-Worker DAGs), ReAct Loops, Model Context Protocol (MCP), FAISS, Vector RAG

Cognitive Systems: Tri-Hybrid RRF ($k = 60$), Knowledge Graphs (BFS Traversal), Dynamic Token Budgeting, Prompt Taxonomy

Frameworks & Storage: PySide6 (Qt), FastAPI, TanStack Start, React 19, Tailwind CSS v4, SQLite (FTS5 WAL), Supabase (PostgreSQL)

Systems & Architecture: Asynchronous I/O (`asyncio`), Multiprocessing (GIL Bypass), IPC, NVML GPU Telemetry, Git, GitHub Actions CI/CD

FLAGSHIP SYSTEMS & ENGINEERING PROJECTS

VISAR EDGE – Autonomous Cognitive OS Layer & Multi-Agent Swarm

2025 – 2026

Core Stack: Python, PySide6, Model Context Protocol (MCP), FAISS, SQLite FTS5, AsyncIO

- Architected a desktop-native cognitive OS overlay powered by a **Master Supervisor Swarm** that dynamically decomposes multi-step user prompts into parent-child task DAGs with in-flight steer injection and isolated worker roles (Coder, Researcher, System Probe).
- Engineered a **5-Tier Cognitive Memory Engine** uniting dynamic token budgeters, SQLite FTS5 ($< 3ms$ BM25 search), semantic entity knowledge graphs, and a **Tri-Hybrid Reciprocal Rank Fusion (RRF, $k = 60$)** synthesizer combining sparse, graph, and dense FAISS vectors.
- Isolated neural wake-word evaluation and vector indexing into dedicated **multiprocessing child workers**, eradicating 100% of Python GIL contention and slashing cold-boot initialization latency from $18s \rightarrow 1.92s$.
- Implemented native **Model Context Protocol (MCP)** stdio/SSE transports with zero-idle lifecycle governors that halt background UI paint loops during window minimization, maintaining an audited $\sim 0.0\%$ **idle CPU utilization**.

LIGHTWEIGHT NLP INTELLIGENCE SUITE – Global Datathon Champion

2026

Core Stack: Python, Scikit-Learn, LightGBM, TF-IDF Sparse Matrices, Pandas

- Awarded **Grand Champion (1st Place)** at the *NeuroLogic '26 Global NLP Datathon*, outperforming heavy LLM baselines using CPU-bound feature engineering, custom ensembling, and zero external API dependencies.
- Trained probabilistic **Soft-Voting Ensembles** (LightGBM, Logistic Regression, Naive Bayes) with Character N-grams (2–5), securing an audited 0.9855 **ROC-AUC** on multilingual code-mixed Hinglish/English toxicity classification.
- Built 5-Fold Stratified Cross-Validation pipelines achieving 99.81% **accuracy** on misinformation detection and 0.8341 Macro F1 on class-imbalanced disaster event filtering under strict sub-second inference constraints.

VISAR INTEL – High-Throughput Automated Semantic News Platform

2026

Core Stack: TanStack Start, React 19, Supabase (PostgreSQL), Groq Cloud API, GitHub Actions CRON

- Engineered an automated current affairs intelligence platform filtering public national RSS feeds through a 2-stage pipeline: a sub-millisecond regex gatekeeper followed by structured JSON LLM summarization.
- Automated scheduled data ingestion via custom GitHub Actions CRON workflows, eliminating serverless execution timeout barriers and populating a custom PostgreSQL database with indexed search vectors.
- Eradicated severe N+1 database query bottlenecks by engineering batched relational queries, paired with a fluid CSS masonry grid and progressive service workers (PWA) enabling complete offline transit reading.

ENGINEERING EXPERIENCE

Applied AI Systems Developer

2024 – Present

Independent Research & Open-Source Infrastructure

- Authored and maintained 40+ modular production repositories specializing in system-level telemetry, low-latency agent loops, and desktop tooling.
- Engineered cross-platform forensic telemetry tools parsing kernel virtual filesystems (`/proc/meminfo`, `dumpsys`), Windows PDH, and NVIDIA NVML for real-time hardware diagnostics without main-thread blocking.
- Designed secure process-isolated sandbox runners executing untrusted subprocess code with hard 15-second timeouts, payload sanitization, and Level-2 command approval gates.

EDUCATION, HONORS & CERTIFICATIONS

Bachelor of Technology – Computer Science (AI & ML Specialization)

2024 – 2028

Gyan Ganga Institute of Technology and Sciences (GGITS) | Jabalpur, Madhya Pradesh

- Honors & Competitions:** Grand Champion (1st Place) – NeuroLogic '26 Global NLP Datathon
- Certifications:** NVIDIA Deep Learning Institute (Fundamentals), Python Essentials, C++ Essentials, IT Essentials